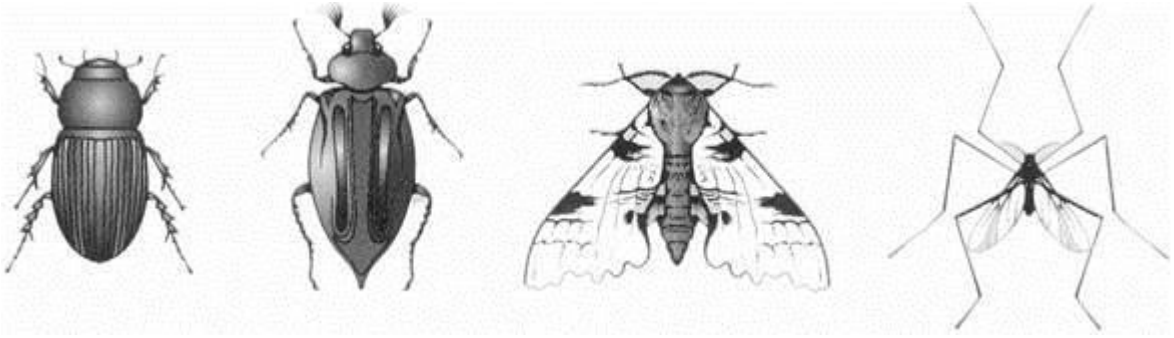
**Q1.** The drawing shows a horseshoe bat.



In an investigation, Val counted the number of insects a horseshoe bat ate.
Her results are shown below.



**A**
35 dung beetles

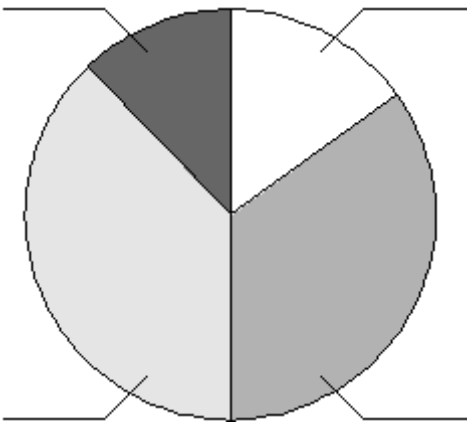
**B**
10 cockchafers

**C**
40 moths

**D**
15 other insects

*not to scale*

(a) Val used the numbers of insects to draw a pie chart.
Write the letter of each insect by the correct section of the pie chart.



3 marks

(b) Dung beetles eat dung (animal droppings).

(i) Why are there lots of dung beetles in fields where cows are kept?

.....

.....

1 mark

(ii) Why do horseshoe bats live close to fields where cows are kept?

.....

.....

1 mark

(c) (i) Insecticides are used to kill insects. What will happen to the number of horseshoe bats if the insects are killed?

.....

1 mark

(ii) Give **one** other reason why the number of horseshoe bats might change.

.....

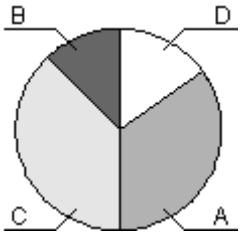
.....

1 mark

Maximum 7 marks

## Q1.

(a)




*if all four answers are correct, award three marks*
*if two **or** three answers are correct, award two marks*
*if one answer is correct, award one mark*
*accept the names of insects instead of letters*
*accept numbers written by the correct sectors*

**3 (L3)**

(b) (i) any **one** from

- plenty of dung **or** food
- cows produce dung

**1 (L3)**

(ii) plenty of dung beetles **or** food
*accept 'dung beetles will be there' **or** 'they eat dung beetles'*

**1 (L4)**

(c) (i) it will decrease

**1 (L4)**

(ii) any **one** from

- change in climate **or** weather
  *accept suitable examples of a change in climate*
- increase **or** decrease in predators
- disease
- habitat **or** roosting sites destroyed
  *accept 'nesting sites destroyed'*
  *accept 'fewer places to live'*
- increase in food supply
- insecticides not used anymore
- insects become resistant to insecticides
  *accept 'fewer cows' **or** 'less food for insects' **or** 'less dung'*
  *accept 'concentration of insecticide in the food chain'*
  *do **not** accept 'they could die'*

**1 (L4)**

**[7]**